

Geometric Interpretation of Covariance and Correlation

1 Geometric Formulation

Consider n observations for two real-valued variables, represented as vectors in an n -dimensional Hilbert space $\mathbb{R}^n$.

Define the mean-centered vectors $\vec{u}$ and $\vec{v}$:

$$\vec{u} = \begin{pmatrix} x_1 - \bar{x} \\ x_2 - \bar{x} \\ \vdots \\ x_n - \bar{x} \end{pmatrix}, \quad \vec{v} = \begin{pmatrix} y_1 - \bar{y} \\ y_2 - \bar{y} \\ \vdots \\ y_n - \bar{y} \end{pmatrix} \quad (1)$$

where $\bar{x}$ and $\bar{y}$ denote the sample means.

1.1 Inner Product and Covariance

The standard Euclidean inner product of $\vec{u}$ and $\vec{v}$ is:

$$\vec{u} \cdot \vec{v} = \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y}) \quad (2)$$

Scaling this inner product by the degrees of freedom $n - 1$ yields the sample covariance:

$$\text{Cov}(X, Y) = \frac{\vec{u} \cdot \vec{v}}{n - 1} \quad (3)$$

Geometric implications:

- $\vec{u} \cdot \vec{v} > 0 \implies$ Parallel alignment (Cov > 0)
- $\vec{u} \cdot \vec{v} < 0 \implies$ Anti-parallel alignment (Cov < 0)
- $\vec{u} \cdot \vec{v} = 0 \implies$ Orthogonality (Cov = 0, linear independence)

1.2 Vector Norms and Correlation

The Euclidean norm of $\vec{u}$ relates directly to the sample standard deviation σ_X :

$$\|\vec{u}\| = \sqrt{\vec{u} \cdot \vec{u}} = \sqrt{\sum_{i=1}^n (x_i - \bar{x})^2} = \sqrt{n-1} \sigma_X \quad (4)$$

The cosine of the angle θ between $\vec{u}$ and $\vec{v}$ is defined as:

$$\cos \theta = \frac{\vec{u} \cdot \vec{v}}{\|\vec{u}\| \|\vec{v}\|} \quad (5)$$

Substituting the norm expressions into Equation (5):

$$\cos \theta = \frac{(n-1)\text{Cov}(X, Y)}{(\sqrt{n-1}\sigma_X)(\sqrt{n-1}\sigma_Y)} = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y} = \rho_{X,Y} \quad (6)$$

By the Cauchy-Schwarz inequality:

$$|\vec{u} \cdot \vec{v}| \leq \|\vec{u}\| \|\vec{v}\| \implies -1 \leq \cos \theta \leq 1 \implies -1 \leq \rho_{X,Y} \leq 1 \quad (7)$$

Summary:

- **Covariance** measures the scaled inner product. It depends on the magnitudes (units) of the vectors.
- **Pearson Correlation** (ρ) measures the cosine of the angle θ between the mean-centered data vectors, normalizing for magnitude and rendering the metric dimensionless.

2 Example

Consider $n = 3$ observational points for variables X and Y :

$$\begin{aligned} X &= [1, 2, 3] \implies \bar{x} = 2 \\ Y &= [2, 4, 6] \implies \bar{y} = 4 \end{aligned}$$

The mean-centered vectors are:

$$\vec{u} = \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix}, \quad \vec{v} = \begin{pmatrix} -2 \\ 0 \\ 2 \end{pmatrix} \quad (8)$$

2.1 Covariance Computation

$$\text{Cov}(X, Y) = \frac{\vec{u} \cdot \vec{v}}{3-1} = \frac{(-1)(-2) + (0)(0) + (1)(2)}{2} = \frac{4}{2} = 2 \quad (9)$$

2.2 Correlation Computation

Calculate the norms:

$$\|\vec{u}\| = \sqrt{(-1)^2 + 0^2 + 1^2} = \sqrt{2} \quad (10)$$

$$\|\vec{v}\| = \sqrt{(-2)^2 + 0^2 + 2^2} = 2\sqrt{2} \quad (11)$$

Computing ρ :

$$\rho_{X,Y} = \cos \theta = \frac{\vec{u} \cdot \vec{v}}{\|\vec{u}\| \|\vec{v}\|} = \frac{4}{\sqrt{2} \cdot 2\sqrt{2}} = \frac{4}{4} = 1 \quad (\theta = 0^\circ) \quad (12)$$

Since $\vec{v} = 2\vec{u}$, the vectors are strictly collinear ($\theta = 0^\circ$), resulting in maximal linear correlation ($\rho = 1$).